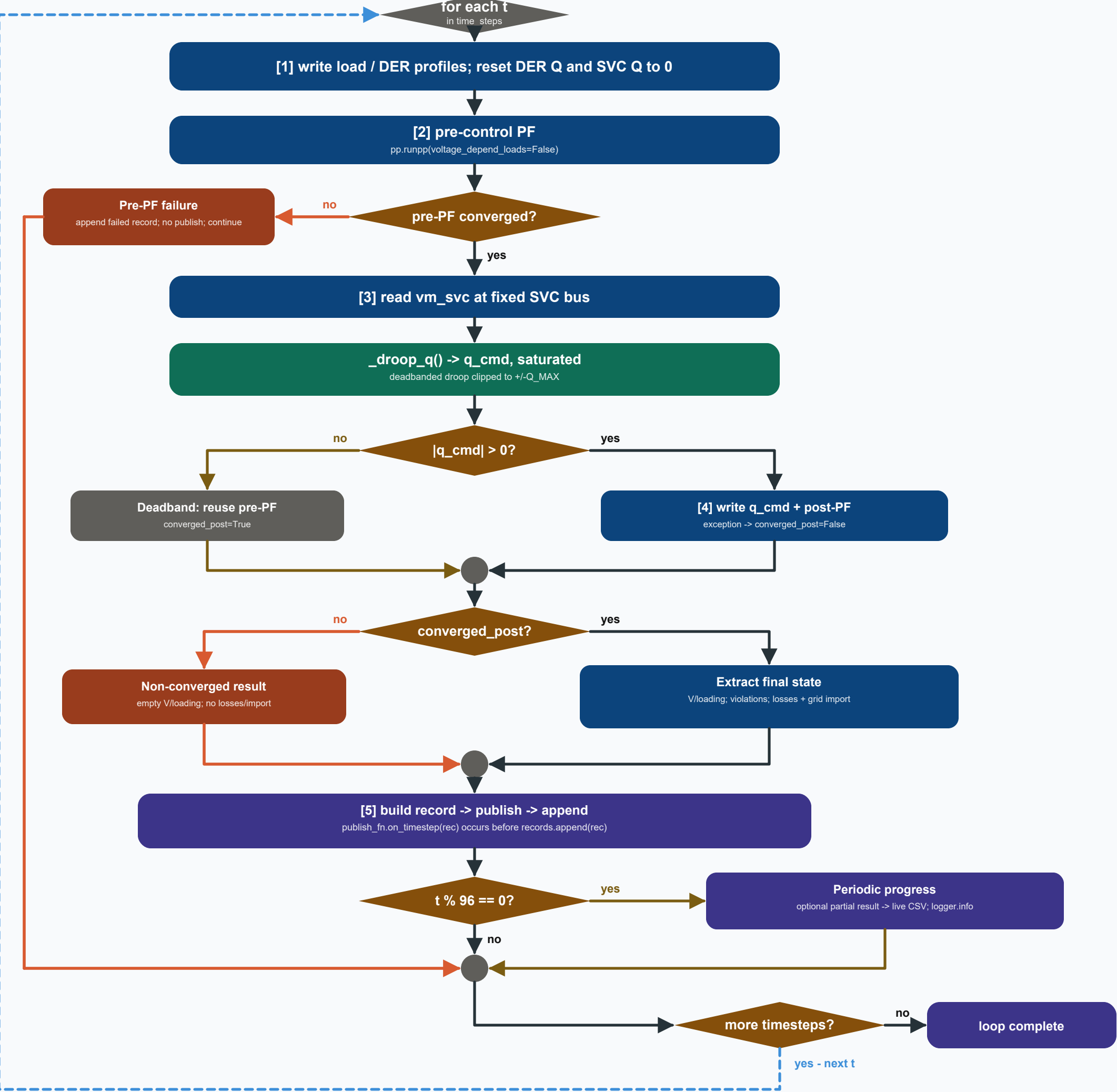




Deadbanded droop law

$\text{error} = 1.00 - \text{vm_svc}$; deadband = 0.01 pu.

$|\text{error}| \leq 0.01 \rightarrow \text{q_cmd}=0, \text{saturated}=\text{False}$.

Undervoltage: $\text{q_raw} = k_q \cdot (\text{error} - 0.01)$, positive Q injection.

Overvoltage: $\text{q_raw} = k_q \cdot (\text{error} + 0.01)$, negative Q absorption.

$\text{q_cmd} = \text{clip}(\text{q_raw}, -Q_MAX, +Q_MAX)$; saturated when $|\text{q_raw}| \geq Q_MAX$.

$Q_MAX = 0.20 \cdot \text{sum}(\text{trafo sn_mva})$; $k_q = Q_MAX / 0.03$.

PF and result semantics

Pre-PF always starts with DER Q=0 and SVC Q=0.

$\text{q_cmd}=0$ skips the second runpp and reuses the valid pre-PF result tables.

A post-PF exception does not abort the scenario; the timestep is recorded converged=False.

Converged records threshold V, line loading, and trafo loading directly from result tables.

There is no active-P curtailment and no DER reactive-power control in Scenario 3.

Publisher / failure semantics and legend

Normal path: build rec -> publish_fn.on_timestep(rec) -> records.append(rec).

PublishHandle checkpoints every normal-path timestep; dashboard live frames use update_every_k.

Pre-PF failure path is different: records.append(failed_record) then continue.

Therefore pre-PF failures skip on_timestep(), checkpoint/live publishing, and the t%96 branch.

The red failure bypass rail and the blue next-timestep loopback rail are separate.

After the loop, top-level finally removes the temporary SVC before final aggregation.

